Factorized RLCA: Action Factorization for Sample-Efficient Multi-Joint Control

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1 Project Overview

1.1 Profile

Track: Algorithm

Abstract and project goal: This project proposes a hybrid algorithmic framework, Factorized-RLCA(Factorized Reinforcement Learning with Combinatorial Actions), designed to bridge the gap between high-dimensional continuous robotic control and large-scale combinatorial optimization. Standard Reinforcement Learning typically treats action spaces as flat vectors, which leads to the Curse of Dimensionality in complex environments. Our primary goal is to develop an efficient framework that decomposes complex joint actions into joint-specific sub-policies to exploit morphological symmetry Ordonez-Apraez et al. [2023]. This approach aims to leverage action space discretization Tang and Agrawal [2020] to significantly reduce the computational complexity and search overhead typically associated with continuous control, enabling accelerated convergence while achieving training results comparable to high-dimensional continuous methods.

1.2 Motivation

The integration of Reinforcement Learning (RL) into multi-jointed robotic systems, such as the `Hopper-v5`, represents a major frontier in autonomous control. However, standard state-of-the-art frameworks typically rely on continuous, policy-based methods (e.g., SAC and PPO) that directly output floating-point control vectors (e.g., continuous joint torques). While intuitive, this approach suffers from two severe limitations that hinder practical deployment: the *Curse of Dimensionality* in policy exploration, and the *Prohibitive Computational Overhead* on resource-constrained hardware.

In standard policy-based continuous RL, the agent explores an infinite action space $\mathcal{A} \subset \mathbb{R}^3$. To learn a stable hopping gait, the network must precisely synthesize floating-point values for multiple joints simultaneously. A single sub-optimal continuous torque combination often leads to immediate imbalance and termination, resulting in low sample efficiency and slow convergence. Furthermore, evaluating the expected return of an infinite continuous policy requires expensive approximation via a continuous Critic network.

To overcome these challenges, we propose a paradigm shift from continuous policy-based execution to structured, value-based discrete evaluation. By applying a uniform discretization to each joint's torque range into 11 atomic levels, we transform an unconstrained continuous search space into a manageable, finite grid.

However, a naive transformation leads to a flat action space that explodes exponentially ($11^3 = 1,331$ joint actions), making value estimation computationally heavy. Our core insight is to factorize the global value function into joint-specific sub-policies. Instead of assessing all 1,331 combinations simultaneously, our proposed *Factorized-RLCA* network isolates and evaluates actions per dimension, outputting only $11 \times 3 = 33$ discrete action scores (Q-values) in a decoupled configuration.

This architectural shift yields profound advantages for edge deployment and embedded hardware (e.g., low-power microcontrollers and robotic edge nodes):

- **Elimination of Real-time Floating-point Scaling:** Rather than computing expensive continuous projections or non-linear activations (e.g., `Tanh`) at the output layer to generate exact torques, the embedded controller performs a simple integer `argmax` operation over the discrete branches. The resulting integer action indices are mapped directly to physical control signals via a pre-defined Look-Up Table (LUT), completely removing floating-point generation from the inference output pipeline.
- **Synergy with Integer-only Quantization:** While the current network's internal weights still utilize floating-point precision, transforming the output space into structured discrete tokens establishes a vital foundation for end-to-end edge optimization. As a prominent direction for future work, this discrete value-based architecture is uniquely compatible with ultra-low-bit quantization frameworks, such as BitNet (1-bit or 1.58-bit representations). By replacing internal floating-point weights with ternary or low-bit integer operations, the entire pipeline can eventually be executed using integer-only arithmetic, achieving high-frequency, real-time robotic control on hardware devoid of Dedicated Floating-Point Units (FPUs).

2 Problem Formulation

We model the Hopper-v5 locomotion task as a Markov Decision Process (MDP). The environment consists of a single-legged robot with a 3-dimensional continuous action space $\mathcal{A} \subset \mathbb{R}^3$, where each dimension corresponds to the torque applied to the thigh, leg, and foot joints, respectively. The agent's objective is to develop a stable jumping gait to move forward (positive x-direction) while maintaining balance. The episode terminates immediately if the hopper's torso tilts excessively or falls below a specific height, making precise joint coordination critical for survival.

To address the efficiency of action exploration, we discretize each continuous torque range into 11 atomic levels within the interval $[-1, 1]$. In our implementation, we apply a uniform discretization to the continuous action space of each joint. Specifically, the action range $[-1.0, 1.0]$ is partitioned using a step size of $\Delta a = 0.2$, resulting in 11 discrete levels. The resulting discrete action set $\mathcal{A}_d$ for each joint is defined as:

$$\mathcal{A}_d = \{-1.0, -0.8, -0.6, -0.4, -0.2, 0, 0.2, 0.4, 0.6, 0.8, 1.0\} \tag{1}$$

This discretization transforms the infinite continuous search space into a finite, manageable manifold while preserving sufficient granularity for precise control. While a naive joint discretization results in a flat action space of $11^3 = 1,331$ possible combinations, we propose a Factorized-RLCA approach to restructure this space. Specifically, we evaluate and compare four different factorization architectures to find the optimal balance between coordination and complexity:

- **Full Joint Control (Baseline):** A single policy outputting $11^3 = 1,331$ actions, representing the most complex search space.
- **Partial Factorization (Hierarchical):** Two configurations consisting of $11 + 11^2 = 132$ actions, where one joint is controlled independently while the others remain coupled.
- **Decoupled Factorization (Independent):** A fully factorized structure with $11 + 11 + 11 = 33$ actions, where each joint (thigh, leg, foot) is controlled by an independent sub-policy.

By transforming the $1,331$ action candidates into a decoupled set of 33 or 132 choices, we aim to demonstrate that localized sub-policies can effectively emerge and coordinate to form complex global jumping behaviors with significantly higher sample efficiency and lower computational cost for embedded deployment.

3 Empirical Evaluation

3.1 Performance Metrics

To rigorously evaluate the effectiveness of the Factorized-RLCA framework, we will employ the following performance metrics:

- **Average Episodic Reward:** The primary metric to measure the agent's ability to maximize cumulative rewards in the Hopper-v5 and LunarLander-v3 environments.
- **Convergence Speed:** The number of training steps or environmental interactions required to reach a predefined performance threshold, highlighting the efficiency of our discretization strategy.

3.2 Baseline Methods

We will compare our proposed method against the following standard reinforcement learning baselines:

- **Proximal Policy Optimization (PPO):** A popular on-policy actor-critic method that uses a clipped objective function to ensure stable policy updates.
- **Soft Actor-Critic (SAC):** An off-policy framework that optimizes a maximum entropy objective to encourage exploration and robust learning in continuous spaces.

3.3 Benchmark Tasks or Datasets

Our evaluation will be conducted on two primary benchmark environments from the OpenAI Gym and MuJoCo suites:

- **LunarLander-v3 (Toy Test):** A classic continuous task used as an initial benchmark to verify the framework's ability to handle basic discretization and landing constraints in a low-dimensional setting.
- **Hopper-v5 (Target Task):** A MuJoCo-based environment involving a single-legged robot that must learn to hop forward. This task serves as the primary evaluation for our factorized architecture, as it requires precise coordination of three joints and strict adherence to stability constraints to prevent falling.

4 Methodology

The core idea of Factorized-RLCA is to transform the high-dimensional continuous control problem into a structured, discrete combinatorial optimization task that is both computationally efficient and inherently safe. We achieve this by first applying a uniform discretization to the action space of each individual joint, converting the infinite search manifold into a finite set of control levels. To prevent the combinatorial explosion of actions as degrees of freedom increase, we employ a factorized architecture that decomposes the global policy into localized, joint-specific sub-policies, allowing the agent to exploit the morphological symmetry of the robot. This hybrid approach enables the agent to benefit from the simplified exploration of discrete spaces while maintaining the rigorous safety guarantees of optimization-based methods.

5 Experimental Results

In this section, we present the preliminary results of our empirical evaluation, comparing the proposed Factorized-RLCA against standard baselines (PPO and SAC). The evaluation is divided into a Toy Test (LunarLander-v3) for initial validation and the Target Task (Hopper-v5) for objective performance assessment.

5.1 Evaluation of Toy Test: LunarLander-v3

The Toy Test on LunarLander-v3 was used to evaluate two different action decomposition strategies within the Factorized-RLCA framework, paralleling the proposed design for Hopper-v5. We tested two variants: **V1 (Joint Discretization)**, which maintains a joint action space of $11 \times 11 = 121$ candidates, and **V2 (Additive Factorization)**, which reduces the complexity to $11 + 11 = 22$ candidates through structural decomposition.

As summarized in Table 1, the experimental results demonstrate a clear trade-off between baseline approximation and sample efficiency. Factorized-RLCA V1 achieved a high final reward of 270 at 654460 steps, closely approaching the asymptotic performance of continuous SAC (275) and outperforming continuous PPO (250). Crucially, Factorized-RLCA V2, which utilizes the most aggressive space reduction, accelerated the convergence to 506264 steps while maintaining a competitive reward of 236. Although V2 exhibits a minor drop in peak reward due to the loss of cross-axis joint information, its significantly smaller action manifold (22 vs. 121 nodes) provides substantially better sample efficiency. This superior scalability-to-performance ratio validates the theoretical foundation of structural decomposition, which is critical for mitigating the curse of dimensionality in the target Hopper-v5 task.

Table 1: Comparative Results for LunarLander-v3 (Toy Test)

Method	Action Space	Steps	Max Reward
PPO (Continuous)	∞	500,000	250
SAC (Continuous)	∞	400,000	275
Factorized-RLCA (V1)	11×11 (121)	654,460	270
Factorized-RLCA (V2)	$11 + 11$ (22)	506,264	236

5.2 Evaluation of Control Configurations: Hopper-v5

To evaluate the impact of factorization on high-dimensional control, we analyze three distinct action configurations for the Hopper-v5 environment. The baseline results for standard continuous methods (PPO and SAC) alongside our proposed configurations are summarized in Table 2.

Table 2: Baseline and Factorized-RLCA Performance on Hopper-v5

Method / Configuration	Action Space Size	Steps	Final Reward
PPO (Continuous)	∞	401,452	3,273
SAC (Continuous)	∞	450,000	3,085
Full Joint Control	$11^3 = 1,331$	652,973	2,714
Partial Factorization	$11 + 11^2 = 132$		
Coupled Thigh-Ankle		703,516	3,149
Coupled Calf-Ankle		**301,173**	**2,828**
Coupled Thigh-Calf		653,206	2,725
Decoupled Factorization	$11 + 11 + 11 = 33$	351,425	2,633

Our Hopper-v5 results validate the central premise of Factorized-RLCA: a carefully chosen factorization can match the performance of continuous policy-based control while training substantially faster. The decisive factor is which joints are coupled, and the standout configuration is **Coupled Calf–Ankle** — the variant that decouples the thigh into its own independent sub-policy while keeping the lower calf–ankle pair jointly controlled.

The convergence of this configuration is the fastest of every method we evaluated — roughly 25%–33% fewer interactions than the PPO and SAC baselines, and less than half the steps required by the Full Joint Control model. It reaches a competitive final reward of 2828 while collapsing the action space from 1331 flat combinations to just 132 structured candidates. The result is intuitive from a morphological standpoint: the thigh drives the global hopping rhythm and benefits from an independent policy, whereas the calf and ankle act as a tightly coordinated lower-limb unit whose coupling preserves the fine balance needed for stable landings.

By contrast, coupling the thigh with another joint (**Coupled Thigh–Ankle** and **Coupled Thigh–Calf**) inflates training to over 650000–700000 steps, and the fully **Decoupled Factorization** (33-action) variant, while compact, sacrifices the cross-joint coordination that the calf–ankle pairing retains.

Together, these findings demonstrate that Factorized-RLCA does not merely shrink the action space—it lets us exploit the robot's morphological structure to achieve baseline-level control with markedly higher sample efficiency, exactly the efficiency-oriented behavior our framework was designed to deliver for resource-constrained deployment.

6 Expected Contributions of Each Team Member

- **Ming-Wei Hsiao**: Focuses on the implementation and evaluation of the proposed framework. His tasks involve developing and training the Factorized-RLCA models across both LunarLander-v3 and Hopper-v5 environments. He is responsible for executing hyperparameter tuning and conducting comparative experiments across various discrete action configurations to validate the effectiveness of action space reduction.
- **Yin-Fei Jiang**: Serves as the project coordinator and lead researcher. Primary responsibilities include conducting the literature review and state-of-the-art survey, formulating the methodology for the factorized RLCA framework, and designing the implementation roadmap. Additionally, she is responsible for data synthesis, results analysis, and the final composition of the research report.
- **Chia-Yu Chang**: Manages the baseline evaluation and benchmarking process. His core responsibility is to implement and train the baseline models, including Proximal Policy Optimization (PPO) and Soft Actor-Critic (SAC), for both the toy test and the objective task. These results provide the essential performance benchmarks required to assess the relative efficiency and convergence improvements of our proposed method.

References

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